11. Sequences and Series

11.11. Applications of Taylor Polynomials

11.11 Learning Objectives
- Understand how to use Taylor Polynomials to estimate functions, limits, and more.
- Understand Taylor's inequality to discuss the error when approximating functions with Taylor Series.

11.11.1. Approximating Functions with Taylor Polynomials

The practically applicable part of this section is approximating the series with n^{th} partial sums, or Taylor Polynomials.

Theorem 11.11.1

The **truncated Taylor series** is a Taylor polynomial of degree n, T_n. The nth degree Taylor plynomial centered at $x = a$ is given by

$$\begin{aligned} T_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \\ &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n \end{aligned}$$

The remainder, $R_n(x)$, of this Taylor series is defined by

$$R_n(x) = f(x) - T_n(x)$$

If $f(x) = T_n(x) + R_n(x)$ and

$$\lim_{n\to\infty} R_n(x) = 0$$

for $|x-a| < R$, then f is equal to the sum of its Taylor series on the interval $|x-a| < R$.

In section 11.10 we were assuming that $\lim_{n\to\infty} R_n(x) = 0$.

Long story short, functions don't have to match their Taylor series representations. Most the functions we will see, will match their Taylor series representation, but we need to be aware that this is not always the case. For example, the following function does not match its Taylor series.

Exercise 11.11.1

Consider the function

$$f(x) = \begin{cases} e^{-\frac{1}{x}} \text{ if } x > 0 \\ 0 \text{ if } x \leq 0 \end{cases}$$

f has continuous derivatives of any order at every real value of x. The derivatives are given by

$$f^{(n)}(x) = \begin{cases} \frac{p_n(x)}{x^{2n}} f(x) & \text{if } \ x > 0 \\ 0 & \text{if } \ x \leq 0 \end{cases}$$

where the p_n is a polynomial of degree $n-1$ that is not relevant to this problem. You can assume that the derivatives are continuous at $x = 0$, and if you want to go through this in office hours that is fine with me.

However, the Taylor series approximation of f does not equal f. To see this, find the Taylor series expansion for f centered at $x = 0$, and graph $f(x)$. Are they the same? (The Taylor series expansion should be *very* easy to graph, if it isn't, you probably made a mistake.)

Warning

Talking about functions that don't match their Taylor series expansions starts to get into some very interesting but very complicated math. Don't stress too much if this is weird. We have to get some particularly strange functions to make them not match their Taylor series expansion. If you are a math major, this starts to get into some very fun topics.

And remember, you won't be tested on 11.11 material, but this is the coolest/most important topic in calculus. It might be worth understanding what makes this all so cool.

Here's a more practical example.

Example 11.11.1

Historically, there have been many ways to calculate digits of π. Since the 14th century, the most prominent method has been utilizing power series. In the 14th century, an Indian mathematician named Madhava correctly approximated π to 11 decimal places. Use the Taylor series representation for $f(x) = \arctan(x)$ to approximate π to 11 decimal places. How many terms of the series are needed to get this accurate of an approximation? (How would you like to do this calculation without a calculator to 11 digits of accuracy?)

Solution

We have shown that $\arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$ for $x \ \text{ in } \ (-1, 1]$. This means, since $\arctan\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$ and $\frac{\sqrt{3}}{3} \approx 0.57 \ \text{ is in } \ (-1, 1]$ that

$$\begin{aligned}\pi &= 6\arctan\left(\frac{\sqrt{3}}{3}\right)\\ &= 6\sum_{n=0}^{\infty}(-1)^n\frac{\left(\frac{\sqrt{3}}{3}\right)^{2n+1}}{2n+1}\\ &= \frac{6\sqrt{3}}{3}\sum_{n=0}^{\infty}(-1)^n\frac{\left(\frac{1}{\sqrt{3}}\right)^{2n}}{2n+1}\\ &= \left(2\sqrt{3}\right)\sum_{n=0}^{\infty}\frac{(-1)^n}{3^n(2n+1)}\end{aligned}$$

This is an alternating series, so using the alternating series estimation theorem, to get 11 digits correct we want

$$\begin{aligned}b_{n+1} = \frac{2\sqrt{3}}{3^{n+1}(2n+3)} &< \frac{1}{10^{11}}\\ 2\sqrt{3}(10^{11}) &< 3^{n+1}(2n+3).\end{aligned}$$

Checking a few values of n we can (eventually) see that $n = 20$ this is true. Starting at $n = 0$ we need to add 21 terms to approximate π to 11 decimal places. Specifically we need to do

$$2\sqrt{3}\sum_{n=0}^{20}\frac{(-1)^n}{3^n(2n+1)} \approx 3.14159265359$$

to get 11 correct digits.

Note

If you were to use $\arctan(1) = \frac{\pi}{4} = \sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}$ instead, it would take about 200 trillion terms to get this level of accuracy. This highlights the importance of having x being close to the center of the Taylor series.

This is just using the alternating series estimation theorem, but can we be more specific for Taylor series what the error will be?

Theorem 11.11.2 Taylor's Inequality

If $f^{(n+1)}(x) \leq M$ for $|x-a| < d$, then the remainder $R_n(x)$ of the Taylor series satisfies the inequality

$$R_n(x) \leq \frac{M}{(n+1)!}|x-a|^{n+1}$$

Note

- The expression for the remainder is related to the next term in the Taylor series expansion! This is a very cool result.
- The error depends on a bound for the derivatives of f near $x = a$

- The error also depends on how close x is to a, the closer x is to a, the lower the error.

Example 11.11.2

Prove

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

for all x in $(-\infty, \infty)$.

Solution

We need to show that $\lim_{n\to\infty} R_n(x) = 0$, where $R_n(x) \leq \frac{M}{(n+1)!}|x-a|^{n+1}$. The function is $f(x) = \sin(x)$, $a = 0$, and

$$\begin{aligned} f(0) &= \sin(0) = 0 \\ f'(0) &= \cos(0) = 1 \\ f''(0) &= -\sin(0) = 0 \\ f'''(0) &= -\cos(0) = -1 \\ &\dots \end{aligned}$$

So $-1 \leq f^{(n)}(0) \leq 1 = M$ We know f is equal to its Taylor series on the interval $|x| < R$ if $R_n(x) \to 0$. Then, we will use squeeze theorem (for limits) since

$$0 \leq R_n(x) \leq \frac{M}{(n+1)!}\ |x|^{n+1}$$

and $\lim_{n\to\infty} 0 = 0$ and for $|x| < d$,

$$\lim_{n\to\infty} \frac{M}{(n+1)!}\ |x|^{n+1} = \lim_{n\to\infty} \frac{1}{(n+1)!}\ |x|^{n+1}$$

Actually finding this limit is difficult, but if we consider the series $\sum_{n=0}^{\infty} \frac{1}{(n+1)!}\ |x|^{n+1}$, we can test if the *series* converges using ratio test

$$\lim_{n\to\infty} \left| \frac{|x|^{n+2}}{(n+2)!} \frac{(n+1)!}{|x|^{n+1}} \right| = \lim_{n\to\infty} \frac{|x|}{n+2} = 0.$$

So *the series* converges no matter what x is. By the divergence theorem, because the series is absolutely convergent,

$$\lim_{n\to\infty} \frac{1}{(n+1)!}\ |x|^{n+1} = 0.$$

Then, by squeeze theorem, $\lim_{n\to\infty} R_n(x) = 0$.

Since $\lim_{n\to\infty} R_n(x) = 0$, the function is equal to its Taylor series everywhere.

11.11a Section Summary:

We got a bit math-y with this one, but the main take-aways are:

- A function is not always equal to its Taylor (or Maclaurin) series expansion.
- Most the functions we will encounter in this class are equal to their Taylor series, but we can investigate an example of a function that is not equal to its Taylor series expansion.
- We used Taylor polynomials (truncated Taylor series) to approximate π, but we can use other Taylor polynomials to approximate values of other "weird" (transcendental) functions (like trig functions).
- We can use error estimation theorems to bound the error when we use a Taylor polynomial to approximate a function.